Financial Evaluation of RosterLab's Global Logistics Expansion

BUSINFO 716 – Business Analytics for FinTech

Introduction

RosterLab, a New Zealand-based AI rostering startup, is evaluating a five-year strategic expansion into the global logistics industry. The project requires an initial capital investment of NZ$10 million in Year 0 and an additional NZ$1 million system upgrade in Year 3. In addition to the NZ$10 million capital expenditure, an initial marketing and launch expenditure of NZ$0.6 million is incurred at Year 0, resulting in a total upfront cash outflow of NZ$10.6 million. Revenue will be generated by capturing market share in a NZ$2 billion logistics market growing at 2% annually. Market share is expected to increase from 1% in Year 1 to 3% in Year 5.

This analysis evaluates whether the expansion creates shareholder value by estimating the cost of capital, projecting free cash flows, and assessing risk through sensitivity analysis.

Estimating Equity Beta and Cost of Capital

Estimating Comparable Firm Equity Beta

RosterLab is privately held and therefore does not have observable market return data. In finance, when valuing a private firm, systematic risk is estimated using a publicly listed comparable company. TechnologyOne (ASX: TNE), a listed enterprise software firm, was selected because it operates in a technology-driven service industry similar to RosterLab.

Monthly stock returns for TNE and the ASX index were calculated using:$R_t = \frac{P_t}{P_{t-1}} - 1$
Equity beta was estimated using ordinary least squares (OLS) regression:$R_i = \alpha + \beta R_m + \varepsilon$
or equivalently:

$$\beta = \frac{Cov(R_i, R_m)}{Var(R_m)}$$

Beta measures systematic risk, or sensitivity to overall market movements. The regression produced:

$$\beta_{TNE} = 0.659$$

The beta coefficient is statistically significant (p = 0.0047). The R^2 of 12.98% indicates that approximately 13% of return variation is explained by market movements. *(ref: Appendix 1.1)*

The regression produced β = 0.659, indicating moderate systematic risk. Since β < 1, TechnologyOne is less volatile than the overall market. This beta serves as the basis for estimating RosterLab's required return under the CAPM

Unlevering the Beta

Observed equity beta reflects both business risk and financial risk due to debt. To isolate pure operating risk, beta must be "unlevered" using TechnologyOne's capital structure.

TechnologyOne's capital structure: Debt = 12%, Equity = 88%, Corporate tax rate (Australia) = 30%

Debt-to-equity ratio: $D/E = 0.12/0.88 = 0.136$

$$\beta_{asset} = \frac{\beta_{equity}}{1 + (1-T)\frac{D}{E}}$$

Using: β_equity = 0.659, Australian tax rate = 30%, D/E = 12% / 88% = 0.136

$$\beta_{asset} = 0.602$$

Asset beta represents business risk independent of financing structure, RosterLab's capital structure differs from TechnologyOne's.

Re-levering for RosterLab

The asset beta obtained from TechnologyOne (0.602) represents pure operating risk, independent of capital structure. However, RosterLab does not intend to adopt TechnologyOne's financing structure. Instead, the company targets a debt-to-equity ratio of 0.50 for the proposed expansion.

RosterLab targets a D/E ratio of 0.50. Re-levering using NZ tax rate (28%):

$$\beta_{RosterLab} = \beta_{asset}\left[1 + (1 - T)\frac{D}{E}\right]$$
$$\beta_{RosterLab} = 0.82$$

The relevered beta (0.82) reflects RosterLab's target financial leverage and is used to calculate the cost of equity for the WACC and DCF valuation.

Cost of Equity (CAPM)

The required rate of return for equity investors is estimated using the Capital Asset Pricing Model (CAPM).

$$r_E = r_f + \beta(r_M - r_f)$$

CAPM states that the expected return on equity equals the risk-free rate plus compensation for systematic risk. The term $(r_M - r_f)$represents the market risk premium, which compensates investors for bearing non-diversifiable risk. Where Risk-free rate = 4.5% and Expected market return = 12%

$$r_E = 10.64\%$$

The cost of equity (10.64%) represents the minimum return required by shareholders given the project's systematic risk and serves as a key input in the WACC calculation.

Weighted Average Cost of Capital (WACC)

While the cost of equity reflects returns required by shareholders, RosterLab's expansion will be financed using both equity and debt. Therefore, the appropriate discount rate must reflect the weighted cost of both financing sources. This is calculated after tax using:

$$WACC = w_E r_E + w_D r_D(1 - T)$$

With: Debt weight = 33%, Equity weight = 67% Cost of debt = 10%, and $WACC = 9.49\%$
WACC (9.49%) represents the firm's weighted opportunity cost of capital and is the appropriate discount rate for FCFF because the project is financed with both debt and equity. *(ref: Appendix 1.2)*

Project Cash Flow Estimation and NPV

The DCF framework values a project based on the present value of its future cash flows. Therefore, accurate estimation of operating performance and cash flow timing is critical.

Revenue Forecast

Revenue is the primary driver of project value, as it directly determines operating margins and free cash flow generation. Revenue is calculated as:

$$Revenue = MarketSize \times MarketShare$$

The total addressable market grows at 2% annually, while market share increases from 1% to 3% over five years. Revenue increases from NZ$20 million in Year 1 to NZ$64.9 million in Year 5 as market share expands within the growing logistics sector.

Operating Costs

Operating performance depends not only on revenue but also on cost structure.

Variable costs are assumed to be 85% of revenue, meaning that costs increase proportionally with sales. Fixed costs are NZ$1.5 million annually, and marketing expenses decline over time as brand presence strengthens. Variable costs = 85% of revenue, Fixed costs = NZ$1.5 million annually, and Marketing expenses decline over time.

EBITDA is computed as:$EBITDA = Revenue - Variable - Fixed - Marketing$
Operating costs determine EBIT and directly influence free cash flow generation.

Depreciation

Depreciation is a cost that reduces taxable income but does not appear as an actual cash outflow. The initial NZ$10 million investment is depreciated straight-line over five years (NZ$2 million annually). The NZ$1 million system upgrade in Year 3 is depreciated over Years 4–5 (NZ$0.5 million annually). Depreciation creates a tax shield by reducing taxable income and is added back when computing free cash flow. In this case, Straight-line depreciation is applied:

Initial CAPEX: $10M/5 = 2M$ annually
Year 3 upgrade: $1M/2 = 0.5M$ annually (Years 4–5)

Net Working Capital (NWC)

Net working capital is assumed to equal 1.5% of revenue. As revenue grows, additional working capital is required to support receivables, system deployment, and operational expansion.

$$\Delta NWC = NWC_t - NWC_{t-1}$$

Working capital increases reduce free cash flow in the early years but are fully recovered in Year 5.

Growing firms like RosterLab often require incremental cash investment in operations before revenue is fully realized. Ignoring working capital would overstate project value. Recovering NWC at the end reflects the liquidation of operational investment. Net working capital equals 1.5% of revenue and is fully recovered in Year 5.

Free Cash Flow to Firm (FCFF)

WACC represents the weighted cost of all capital providers, cash flows must be measured before interest payments. Using FCFF ensures consistency between the numerator (cash flows) and the denominator (WACC).

Free cash flow is calculated as: $FCFF = EBIT(1 - T) + Depreciation - CapEx - \Delta NWC$

Taxes are calculated as max(0, EBIT × tax rate), meaning tax is set to zero in loss years, and no loss carry-forward benefit is assumed. Project life is 5 years, hence there is no terminal value assumed.

Net Present Value (NPV)

NPV measures the increase in shareholder wealth resulting from the investment. A positive NPV indicates that projected returns exceed the opportunity cost of capital.

Using the DCF model:$NPV = \sum \frac{FCFF_t}{(1+WACC)^t}$

Discounting at 9.49% yields: $NPV = NZ\$2{,}190{,}500$, Since NPV > 0, the project is expected to create economic value. In practice, NPV is preferred because it directly measures the amount of value created in monetary terms. The project's Internal Rate of Return (IRR) is approximately 15.16%, which exceeds the WACC of 9.49%, indicating that the investment is expected to generate returns above the required rate of return.

Sensitivity Analysis

Sensitivity analysis evaluates how changes in key assumptions affect project value.

The following variables were tested independently by adjusting each variable and keeping the other values constant: Market share ±1%, Variable cost ±5% and WACC ±1%. The absolute impact value is measured as ABS (NPV_up − NPV_down); % impact is (impact ÷ base-case NPV). *(ref: Table 1.1)*

Table 1.1 – Sensitivity findings

Variable	Absolute Impact Value	Impact %	Interpretation of NPV change
Market Share	$ 18,757,064.00	856%	approximately 8.6 times the base NPV
Variable cost	$ 11,773,539.00	537%	approximately 5 times the base NPV
WACC	$ 878,424.00	40%	No major impact

The sensitivity analysis result shows that the market share changes have the largest impact on NPV, Variable Cost changes have the second-largest impact and WACC changes have a comparatively smaller impact.

The sensitivity analysis demonstrates that project value is primarily driven by operating assumptions. A ±1% change in market share generates the largest variation in NPV, confirming that revenue scale and successful market penetration are the dominant determinants of value. Variable cost assumptions have the second-largest impact, indicating that cost control and margin management are critical to sustaining free cash flow. *(ref: Appendix 1.3)*

By contrast, a ±1% change in WACC produces a relatively small effect on NPV. Over the five-year period, valuation is more sensitive towards operating cash flow than the changes in the discount rate, suggesting that execution risk outweighs financing risk in this project.

Conclusion

This analysis evaluated RosterLab's proposed expansion into the global logistics industry using systematic risk estimation, CAPM, WACC, discounted cash flow valuation, IRR, and sensitivity analysis. Currently, the project generates a positive Net Present Value of approximately NZ$2.19 million and an Internal Rate of Return of 15.16%, both exceeding the firm's weighted average cost of capital of 9.49%.

From a financial perspective, the projected operating cash flows are sufficient to recover the initial capital losses and generate returns above the required risk-adjusted benchmark. The application of CAPM and WACC ensures that the valuation appropriately incorporates systematic risk and financial leverage, aligning the investment decision with capital market expectations.

The sensitivity results show that the project's value depends mainly on how well the business performs operationally. Even a small 1% change in market share causes a very large change in NPV, much more than changes in costs or the discount rate. Variable costs also have a strong effect, while a 1% change in WACC has only a relatively small impact. This means that the project's success will depend more on achieving sales growth and controlling costs than on changes in financing conditions.

Overall, the expansion is financially workable under the current assumptions. However, because the NPV is sensitive to market share and costs, the company will need to gain customers as planned and manage expenses carefully to achieve the expected value.

References

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Appendix

1.1 Regression Result:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.360217648
R Square	0.129756754
Adjusted R	0.11475256
Standard E	0.082459798
Observatic	60

ANOVA

	df	SS	MS	F	ignificance F
Regressior	1	0.058803	0.058803	8.648032	0.004697
Residual	58	0.394378	0.0068		
Total	59	0.453181			

	Coefficients	andard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.024186848	0.010678	2.265131	0.027256	0.002813	0.045561	0.002813	0.045561
X Variable	0.658923808	0.224066	2.940754	0.004697	0.210406	1.107441	0.210406	1.107441

1.2 WACC calculation

Given Data	Assumption
Tax Rate - NZ	0.28
Tax Rate - AU	0.30
D - TNE	12.00
E - TNE	88.00
Targer D/E	50%
r_F	4.50%
r_M	12.00%
r_D	10.00%

Calculated Data	Value
TNE β	0.659
D/E	0.136
TNE_ASSET β	0.602
Debt Weight	33%
Equity Weight	67%
RosterLab β	**0.82**
r_E (Cost of Equity)	**10.64%**
WACC	**9.49%**

1.3 Sensitivity Findings: